Two-Phase Gaze-Regularized LoRA-Swin Architecture & Loss Framework

Two-Phase Training Protocol

Phase 1: OralXrays-9 Pre-training (12,688 OPGs, λ=0, β=0)
Phase 2: Tufts TDD Fine-Tuning (1,000 OPGs, Gaze λ=0.05 vs Ablation λ=0)

Input OPG

Panoramic X-Ray
224 × 224 × 3

Swin-Tiny Backbone (Frozen)

27.5M params (ImageNet weights frozen)

Stage 1 (56² × 96)	**LoRA (q, v) r=8, α=16**
Stage 2 (28² × 192)	**LoRA (q, v) r=8, α=16**
Stage 3 (14² × 384)	**LoRA (q, v) r=8, α=16**
Stage 4 (7² × 768)	**LoRA (q, v) r=8, α=16**

Trainable LoRA: 141,312 params (0.51%)

U-Net Style Spatial Decoder

5.08M params (Trainable)

Skip-Fusions (56², 28², 14², 7²)

Transposed Convolutions + Blocks

1×1 Conv → 224 × 224 Logit Map s(h,w)

Classification Head & MIL Blend

Global Head: LayerNorm → Linear 768→256 → GELU → Linear 256→1 (z_glob)

Decision Logit Blend Eq. (1):

$$\hat{z} = z_{glob} + \beta \cdot \max_{(h,w)} s(h,w)$$

Phase 2: β = 0.5 | Phase 1: β = 0

Composite Multi-Task Loss Eq. (2)

$$\mathcal{L}_{total} = \mathcal{L}_{cls} + \lambda(t)\mathcal{L}_{gaze} + w_{aux}\mathcal{L}_{seg} + w_{pseudo}\mathcal{L}_{pseudo}$$

1. Classification Loss ($\mathcal{L}_{cls}$)

Weighted BCE + Label Smoothing (ε=0.05)
Applied to decision logit $\hat{z}$

2. Gaze Alignment Loss ($\mathcal{L}_{gaze}$) [Eq. 3]

$D_{KL}(g \parallel a)$ over 48×48 upsampled Swin attention
Warm-up λ(t) up to 0.05 over 8 epochs [Eq. 4]

3. Dense Auxiliary Loss ($\mathcal{L}_{seg}$) [Eq. 5]

$0.5 \cdot \mathcal{L}_{BCE} + 0.5 \cdot \mathcal{L}_{Dice}(s, m) \cdot \mathbb{I}(||m||_1 > 0)$
$w_{aux} = 0.1$ on decoder output s

4. Pseudo-Label Consistency ($\mathcal{L}_{pseudo}$)

BCE toward y=0 for confident Phase-1 negatives
Weight $w_{pseudo} = 0.25$ ($p < 0.15$)